

Empirical Proof Record: VALIDATED

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Proof ID: de5b99c337eb26caa8acde0e02aedc9c...

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1. Claim

Under sustained load, the substrate stays cognitively alive on real input: (for every cycle that produced a concept set, $\Phi \geq 0.05$). Empty-concept cycles are excluded — with nothing to integrate, $\Phi=0$ is correct, not a defect; their rate is reported separately. The floor is the lowest- Φ real-input cycle across the run.

- **Operationalisation:** Stream 5 passes over 8 substantive stimuli through Kimera’s entity target (Takwin); read the per-cycle “phi” and “concepts”. Partition cycles into real-input (concepts non-empty) and empty-input. $\text{phi_floor} = \min \Phi$ over real-input cycles; the LTL invariant holds iff that $\text{floor} \geq \varphi_floor$. empty_input_rate and the over-all-cycles phi_floor_strict are reported alongside.
- **Threshold:** $\text{phi_floor} \geq 0.05$ phi
- **H0:** H0: $\min \Phi < 0.05$ on real-input cycles — Φ collapses below the floor despite the input producing concepts
- **H1:** H1: $\min \Phi \geq 0.05$ on real-input cycles — the substrate stays cognitively alive whenever it has something to integrate

2. Verdict

VALIDATED — observed 0.6603084741552229

Φ floor 0.6603 over 40 real-input cycles (5 passes $\times$ 8 stimuli); mean Φ 0.6892, band [0.6603, 0.7023], stdev 0.0141; non-collapse rate 100.0%; 0 empty-input cycles excluded (0.0%; strict floor over all cycles 0.6603); 0 cycles produced no Φ ; worst real cycle: stimulus 1 @ cycle 1 = 0.6603; cross-check passed: non-collapse Wilson 95% CI [0.912, 1.000] (confidently alive)

3. Evidence

Pillar	Statistic	Value	95% CI	Library
0.flow.phi_stability	phi_floor	0.6603084741552229	—	ophamin 0.97.0

4. Pre-registration

- Registered at: 2026-05-22T13:14:11.738173+00:00
- Config hash: d9e9ad915c6867382837179e54fdf870...
- Data hash: fc88457a1778b2a25ed9f7930f7315a0...

5. Signature

6a7f415740c42624be15d058ee781d84cafcd7d03f96e3ca8d0173be1c0347c0